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Feasibility Assessment of Utilizing N₂ with CO₂ in Enhanced Gas Recovery and CO₂ Sequestration

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Abstract

A significant number of carbon capture and storage (CCS) projects integrate CO₂ sequestration with enhanced gas production (EGR). Integration of CCS and EGR provides a practical approach to maximize gas production while sustaining environmental goals. While numerous lab experiments explored mixing CO₂ and N₂ to increase gas recovery and reduce CO₂ cost, this study examines the effects of N₂-CO₂ mixture on a field scale and explores the effects on CO₂ storage capacity.

To visualize the effects of the N₂-CO₂ mixture on gas recovery, 3-D simulation models were employed to compare utilizing the mixture in a conventional dry gas reservoir. For sensitivity analysis, numerous scenarios were considered, including injection rates and reservoir properties. The model entails a 10-year simulation period, with production starting 4 years before injection, stopping production once the concentration of CH₄ falls to 0.5. Maximum injection pressure is set at 8,000 psi, avoiding potential CO₂ leakage. This will help in mimicking field conditions to assess the maximum economic recovery and CO₂ storage.

While results found in the literature offer contradictory results, the results highlight the positive influence of N₂ on gas production. Field-scale results showcase that combining N₂ with CO₂ could be effectively utilized as it yields an overall similar recovery to pure CO₂. In fact, in lower permeability reservoirs, N₂ provides higher gas recovery than that of CO₂ by 2-3%. However, in higher permeability reservoirs, the results shift the other way, with CO₂ outperforming N₂. This result is mainly driven by the breakthrough of N₂ and CO₂, where in low permeability, CO₂ breakthrough is faster than N₂, while reversing the behavior in high permeabilities. Nonetheless, CO₂ storage efficiency is the same in both cases, effectively storing more than 90% in both cases. The total amount of sequestered CO₂ remains the highest when injecting CO₂ as the initial volume of CO₂. Therefore, this method can be a feasible option for improving project economics while maintaining the environmental objectives.

Incorporating N₂ with CO₂ in EGR projects provides a novel approach to sequestering CO₂ while generating the highest gas production and cost reduction. In addition to reducing processing costs, injection of N₂-CO₂ can reduce potential corrosion issues in production wells by minimizing CO₂ production. Therefore, including N₂ can facilitate the optimization of CCS and EGR projects.

Introduction

Fueled by the increase in CO₂ emissions, a significant increase in global temperatures poses direct effects on humans, endangering communities and advancing ecological changes. Rising sea levels represent an existential threat to communities living in low altitudes and proximity to the sea (Materson et al., 2025). Moreover, ecological changes can result from affecting animals that require adjustment to higher temperatures to survive or face extinction (Weiskopf et al., 2020). Mitigating such perilous environmental changes represents the focus of many industries, including oil and gas, as it is considered the main contributor in terms of CO₂ emissions.

Despite clean energy-yielding lower CO₂ emissions than fossil fuels, susceptibility to seasonal changes and weather conditions poses a significant challenge to meet the rising energy demand. Meanwhile, hydrocarbons remain a reliable source of energy. Nonetheless, the continued utilization of hydrocarbons will increase CO₂ emissions accordingly.

To keep CO₂ emissions low and subsequently lower the concurrent increase in global temperatures, Carbon Capture and Sequestration (CCS) represents a prominent application to curb the increase in CO₂ emissions. Carbon capture can be achieved through implementing separation technologies in fossil fuel power plants. These technologies include mature separation technologies such as absorption, adsorption, and membrane separation. Despite the maturity of these technologies and having been implemented for more than 30 years, further optimization is needed to capture larger volumes of CO₂.

Moreover, companies enjoy high incentives to implement CCS through the 45Q tax credit. The tax credit can offset the cost of capturing by providing a tax credit of \$50/MT of CO₂ sequestered (IEA, 2020). Policymakers have advocated for more incentives to further encourage the implementation of CCS. Capitalizing on these incentives can increase the involvement of oil and gas companies in implementing CCS.

Perhaps a major setback of CCS projects lies in the negligible returns on investments. Sequestering CO₂ in geologic formations lacks tangible returns. Incentives alone are not enough to encourage implementation, as incentives are vulnerable to economic challenges and funding difficulties (Tollefson, 2015). This is evident in current CCS operations that mostly utilize CO₂ for EOR operations (Yusuf & Ibrahim, 2023). Effective utilization of CO₂ can significantly offset the cost associated with CO₂ capture.

Notwithstanding the effectiveness of CO₂ in EOR, several challenges can impede the implementation of CO₂-EOR. The high capital cost of CO₂ capture and subsequent purification of associated gas impose mounting challenges that require adequate attention and optimization (Kumar et al., 2024). Moreover, the transportation of CO₂ to oil and gas fields requires additional capital expenditure. Moreover, the corrosivity of CO₂ poses a real threat to EOR, as this can directly affect wells and facilities (Fan et al., 2025). Typically, CO₂-EOR is applied in existing fields that can contain old wells, already exhibiting integrity issues (Laumb et al., 2017). Therefore, CO₂ necessitates continuous integrity monitoring, ultimately increasing field operation costs and possibly loss of production due to maintenance shutdown. To mitigate such hindering concerns, incorporating other gases in CCS EOR offers a possible approach.

However, gases other than CO₂ typically exhibit higher MMP, lowering the favorability of utilization in EOR operations (Alfarge et al., 2020). Nonetheless, enhanced gas recovery (EGR) started to garner more attention. Conventional gas fields typically enjoy a high primary recovery rate, lowering the need for secondary and tertiary recovery. Nevertheless, the high demand for natural gas in recent years could potentially drive further investments in EGR. The first known CO₂ EGR in the world is the K12 project in the Netherlands, where produced CO₂ was recycled and injected back into the reservoir (Fan et al., 2025). Following the success of the project, similar projects combining CCS and EGR were undertaken in Australia and Germany (Ali Awan & Din Kirmani, 2025). While miscibility can limit the implementation of N₂ EOR, N₂ can play a more prominent role in EGR than in oil.

Numerous experimental studies have examined the feasibility of utilizing N₂ with CO₂ in EGR. [Liu et al. \(2024\)](#) investigated CO₂, 0.5 CO₂-0.5 N₂ mixture, and pure N₂ by employing NMR technology to explore the effectiveness of each component in displacing CH₄ in large, medium, and small pores. The results yielded a better performance by N₂, sweeping CH₄ molecules in large and small pores compared to mainly large pores in CO₂ injection. Nonetheless, the experiment did not further the investigation with multiple injection rates and varying reservoir parameters such as porosity and permeability. On the other hand, [Zhang et al. \(2022\)](#) explored the effectiveness of N₂ in different parameters. The experiment entailed performing displacement under different compositions, porosity and permeability, injection rates, and displacement direction. Three Berea Sandstone cores were examined with 98.4 md, 216.8 md, and 417 md. The results showed better performance by pure CO₂ across all cores, with increasing recovery in higher permeability. In addition, multiple injection rates and different fluid compositions were also studied using pure CO₂, combinations of CO₂-N₂ fluids, and pure N₂. The results continued to show that CO₂ had the highest recovery of CH₄. The contradictory results of the two experiments encouraged the investigation of utilizing N₂ with CO₂ on a field scale level.

Building on previous studies in this area, this study will help to further assess the feasibility of N₂ with CO₂ in EGR operations in conventional reservoirs, addressing the impact of reservoir properties and injection rates on gas recovery and CO₂ storage. Ultimately, finding a balance between the two objectives will serve as a feasible approach to further CCUS implementation.

Methods and Inputs

To entertain the idea of incorporating N₂ with CO₂ in EGR, a 3-dimensional model was created using CMG-GEM. To simplify the study, the reservoir fluid model generated through CMG-WINPROP consisted of CH₄ only with a homogeneous reservoir. The model entailed two horizontal wells opposite each other, with the injection well situated deeper than the producer. The reservoir data used are presented in [Table 1](#). The relative permeability curves were added using ([Oak et al., 1990](#)) from Berea sandstone and shown in [Figure 2](#).

Table 1—Reservoir Properties and Model Dimensions

Parameter	Value	Unit
Number of Grid Blocks (x*y*z)	83×27×7	-
Model Dimensions	545 × 800 × 130	m
Reservoir Porosity	0.1602	-
Reservoir Permeability Range	10, 100, 216.8, 500	md
Water Saturation, S _w	0.3	-
Reservoir Temperature, T	256	°F
Initial Reservoir Pressure, P	4,700	psi

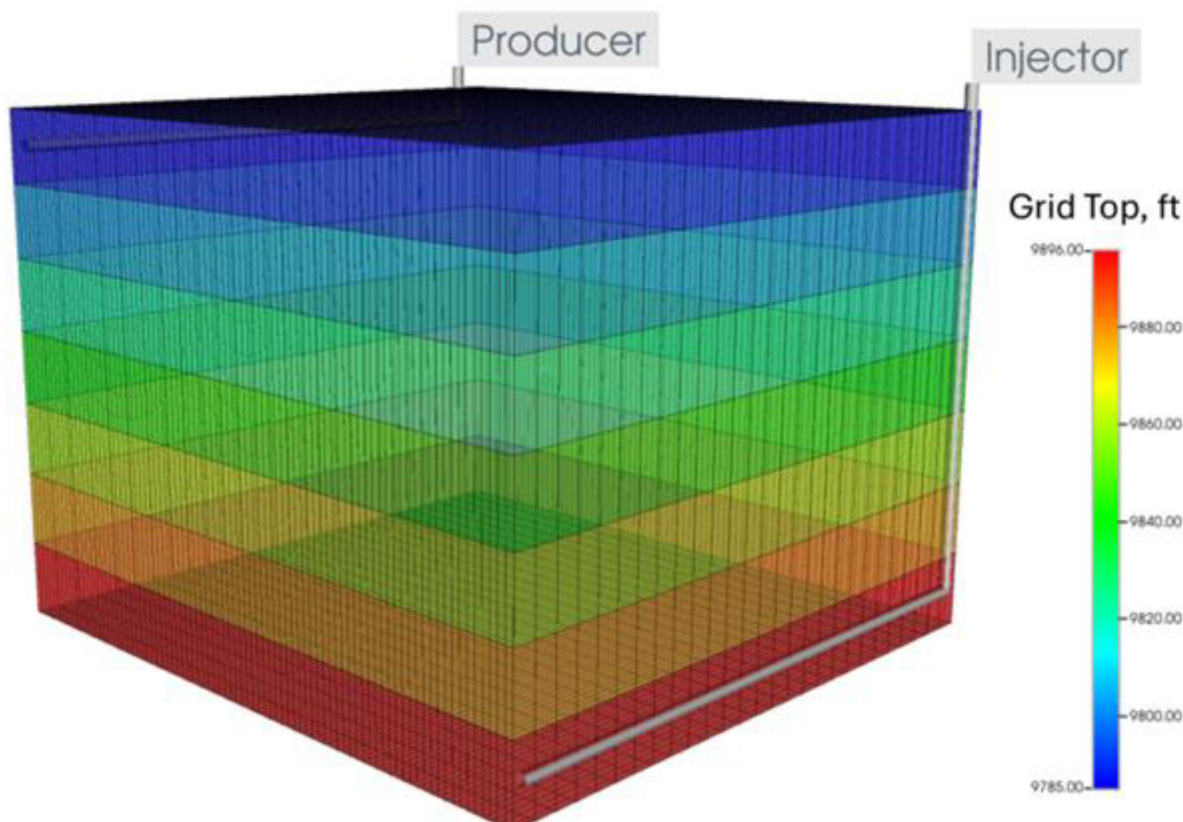


Figure 1—3-D Model of the Reservoir showing the Placement of the Injector and Producer

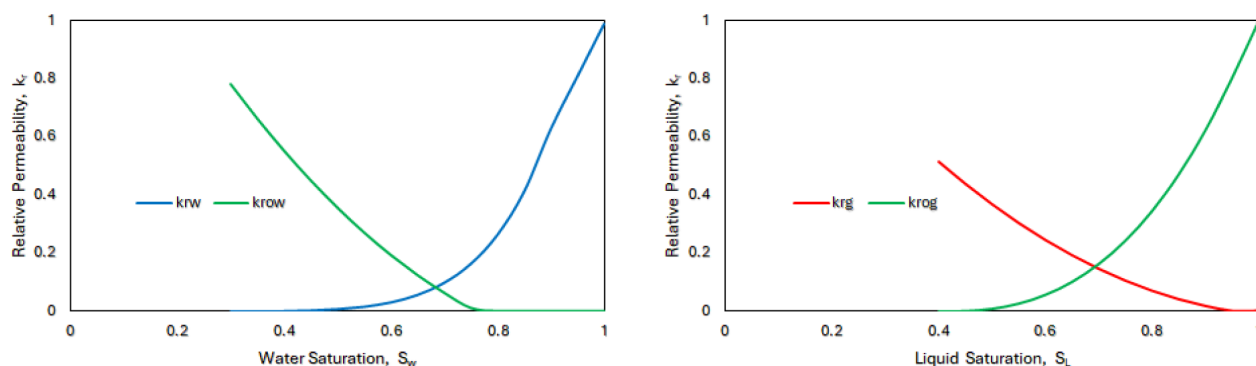


Figure 2—Relative Permeability Curves for Berea Sandstone (Oak et al., 1990)

In addition to the base permeability of 216.8 md, the sensitivity analysis applied included varying permeabilities of 10, 100, and 500 md. Besides reservoir properties, operational conditions, including varying injection rates, were also studied. The injection rates were based on reservoir conditions to compare both fluids to mitigate the influence of pressure on volume, as N_2 and CO_2 volumes depend on pressure. The injection rates include 0.5, 1, 1.5, 2, and 2.5 MRCFD. In addition, the constraint on the producer entailed that the production stops once the concentration of CH_4 reaches a minimum of 0.5, reducing excessive production of both CO_2 and N_2 . To avoid potentially fracturing the reservoir, the maximum injection pressure was set at 8,000 psi. Moreover, the maximum production rate and minimum bottom hole pressure were set at 300 MSCFD and 1,500 psi, respectively. These identified operational parameters can greatly influence

Results and Discussions

The results yielded similar outcomes for N_2 and CO_2 at lower injection rates of 0.5 and 1 MRCFD. However, at 1.5 MRCFD injection rates and higher, N_2 slightly outperforms CO_2 in lower permeability, and the latter outperforms the former in higher permeability. This is caused by N_2 breakthrough faster than CO_2 in higher permeabilities, reducing CH_4 recovery. Whereas in lower permeabilities, CO_2 showed a faster breakthrough than N_2 . Moreover, the higher density of CO_2 accordingly yields higher injection pressure, energizing the reservoir and subsequently increasing CH_4 recovery. Nevertheless, the results illustrate the feasibility of utilizing N_2 in EGR along with CO_2 to maximize recovery in some circumstances.

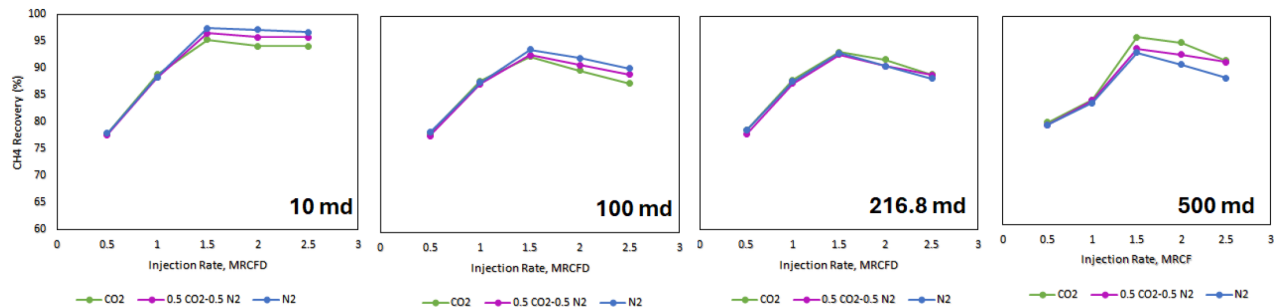


Figure 3—Examining the Influence of Injection Rate and Permeability on the Comparative Advantage of CO_2 and N_2 in CH_4 Recovery

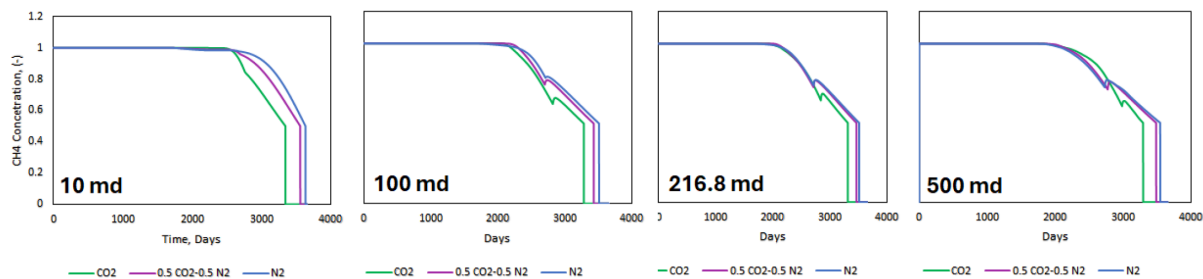


Figure 4—Breakthrough Curves Illustrated by the Decrease of CH_4 concentration in the Produced Gas Stream at an Injection Rate of 1.5 MRCFD

To further illustrate the breakthrough effect, we can compare pure CO_2 and pure N_2 cases. Figure 5 showcases how N_2 breakthroughs develop faster in high permeability compared to lower permeability. Although N_2 appears to break through faster in a 10 md permeability, the severity of the breakthrough is smaller compared to that of CO_2 , where CO_2 concentration increases at a higher rate compared to N_2 .

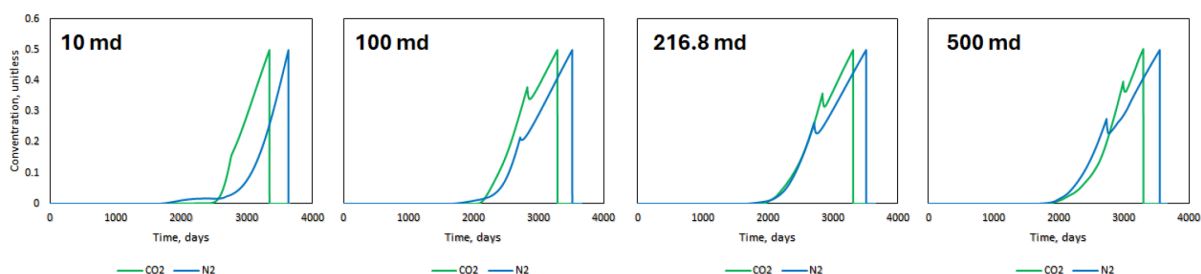


Figure 5—Breakthrough Curves of Pure CO_2 and Pure N_2 across Different Permeabilities at 1.5 MRCFD

Besides the influence of breakthroughs, higher reservoir pressure can overcome fast breakthrough in some cases. At high permeability, the influence of CO_2 on reservoir pressure becomes clear, sustaining gas

rate at higher rates. This in turn increases the recovery. Figure 6 demonstrates how at high permeability, CO₂ managed to generate higher reservoir pressure for a longer time period than pure N₂ and CO₂-N₂ mixture. This shows that at higher permeabilities, CO₂ becomes more favorable than N₂.

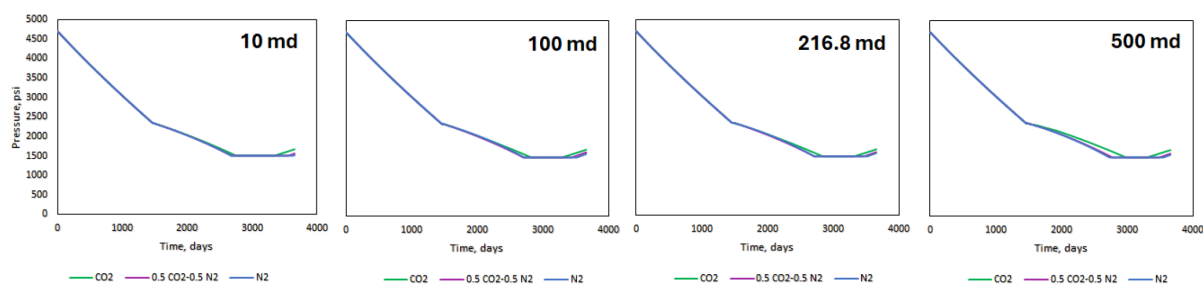


Figure 6—Average Reservoir Pressure Comparison across Different Permeabilities at 1.5 MRCFD

Notwithstanding the impact of both CO₂ and N₂ on CH₄ recovery, the influence on CO₂ storage represents a significant factor in addressing the feasibility of injecting N₂ with CO₂. Although oil and gas proceeds to generate higher revenue, CO₂ storage and CO₂ production also play a vital role in project economics. By injecting CO₂ alone, higher amounts of CO₂ will consequently be produced. As demonstrated Figure 7, the amount of produced CO₂ doubles in most cases. Having a mixture of N₂ and CO₂ reduces the production of CO₂ and lowers the risk of CO₂ corrosion.

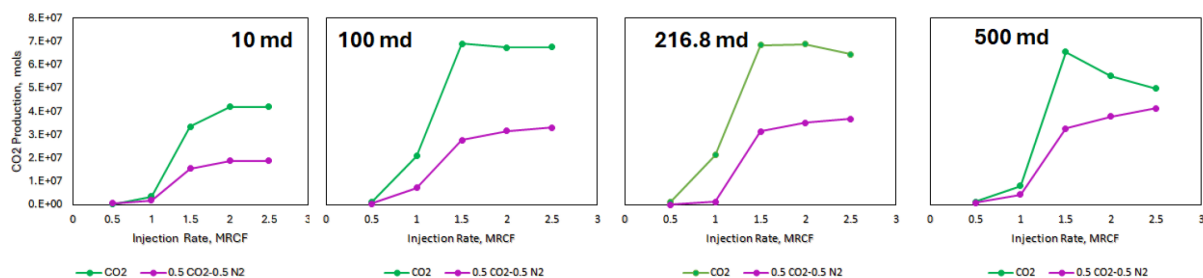


Figure 7—CO₂ Production Across Different Permeabilities for Pure CO₂ and CO₂-N₂ Mixture

While injecting pure CO₂ stores higher amounts of CO₂, storage efficiency remains relatively the same for both pure CO₂ and CO₂-N₂ mixture, with pure CO₂ slightly exhibiting higher efficiency in some cases. Meaning that the amount of stored CO₂ compared to the injected amount remains significantly high. Storage efficiency consistently drops at 1.5 MRCFD across all permeabilities, triggered by the breakthrough of CO₂. However, efficiency increases slightly afterward due to the earlier shutdown of the producer to avoid excess production of CO₂, causing longer periods of CO₂ injection.

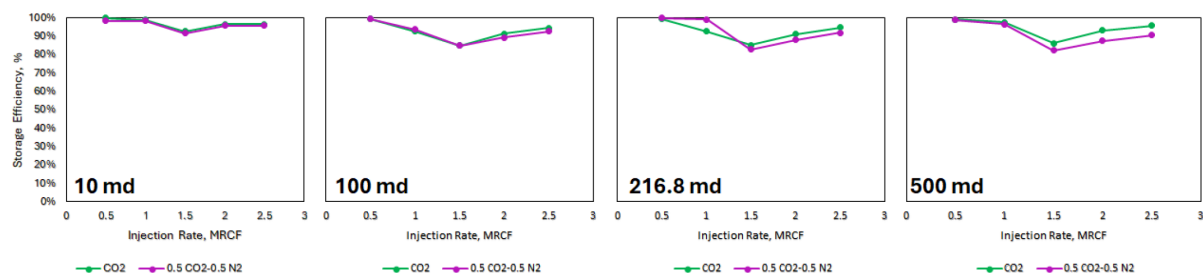


Figure 8—Assessment of CO₂ Storage Efficiency Across Different Permeabilities and Injection Rates

Conclusions

The result of the simulation runs highlights the positive influence of N_2 on CH_4 recovery, affirming the viability of utilizing N_2 with CO_2 in EGR. The study focused on establishing an understanding of how N_2 behaves in EGR on a field-scale level, including the effect on CO_2 storage efficiency. The following summarizes the main findings of the study and highlights areas of future studies:

- The comparative breakthrough times of CO_2 and N_2 across different permeabilities heavily influence the recovery. CO_2 breaks through faster in low permeability. Whereas in high permeability, N_2 breaks through faster than CO_2 . This consequently affects the production of CH_4 , affecting the recovery rate accordingly.
- The influence of pressure becomes more evident at higher permeabilities, favoring CO_2 due to higher density. Nevertheless, N_2 - CO_2 still yields high recovery rates, which can influence the decision-making considering how the presence of N_2 lowers CO_2 production.
- Although the stored amount of CO_2 remains higher with pure CO_2 injection, combining N_2 with CO_2 provides slightly lower storage efficiency, maintaining the environmental goal of CO_2 sequestration.
- N_2 plays a vital role in lowering CO_2 production, as N_2 entails no corrosion and integrity risks. In addition, this also reduces the cost of gas purification from CO_2 residues.
- Future studies could explore the viability of utilizing N_2 in condensate gas reservoirs, focusing on the effect on enhancing condensate recovery.

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Appendix

The simulation included the diffusion coefficient of both CO₂ and N₂ in CH₄. Eq. 1 & 2 entail the calculation for the diffusion coefficients of both CO₂ and N₂, respectively (Shoushtari et al., 2023).

$$D_{CO_2CH_4} = \frac{8.2 \times 10^{-11} T^{1.75}}{P} \quad (1)$$

$$D_{N_2CH_4} = \frac{10.2 \times 10^{-11} T^{1.75}}{P} \quad (2)$$

where, P and T represent the pressure and temperature of the reservoir, respectively.